

의료와 데이터사이언스 11 주차

(비정형 데이터 생체신호 분석 3: 생체신호AI 모델링2)

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VitalDB

Hands-on: Low blood pressure prediction

Hands-on: Stroke volume prediction

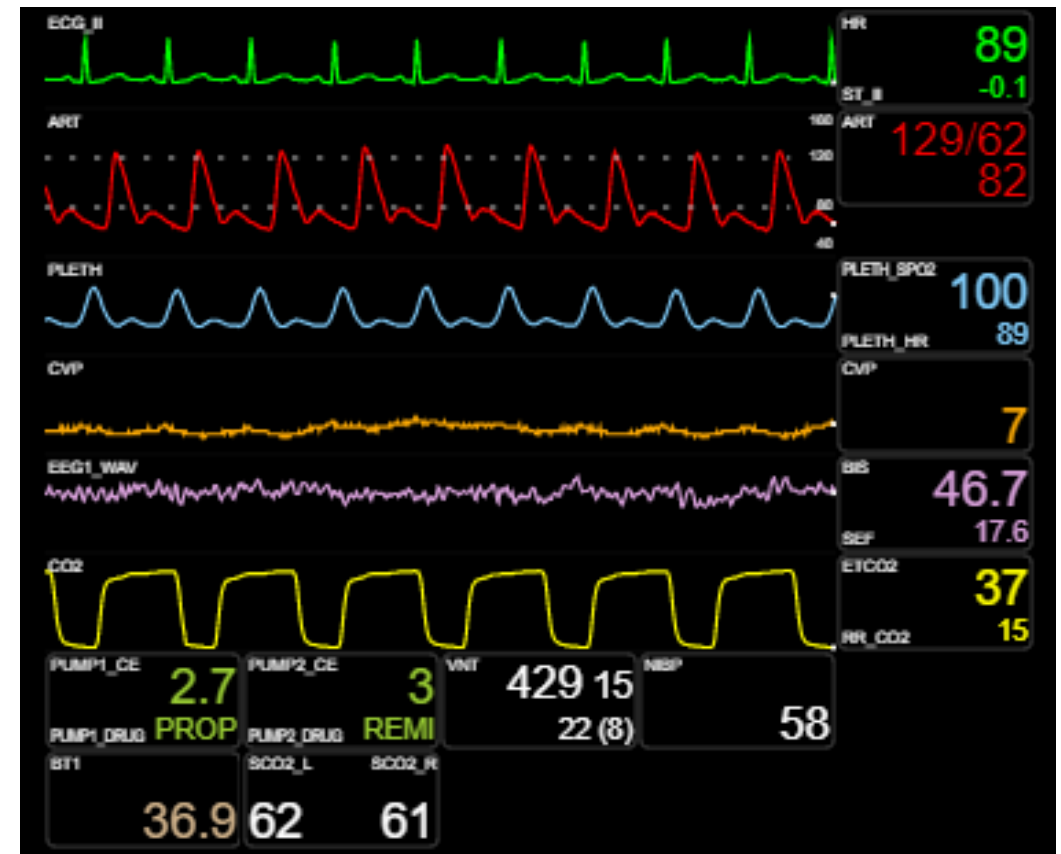
Physiological Signal

● Definition

- electrical, mechanical, thermal, or other signals measured over the time from the human body.

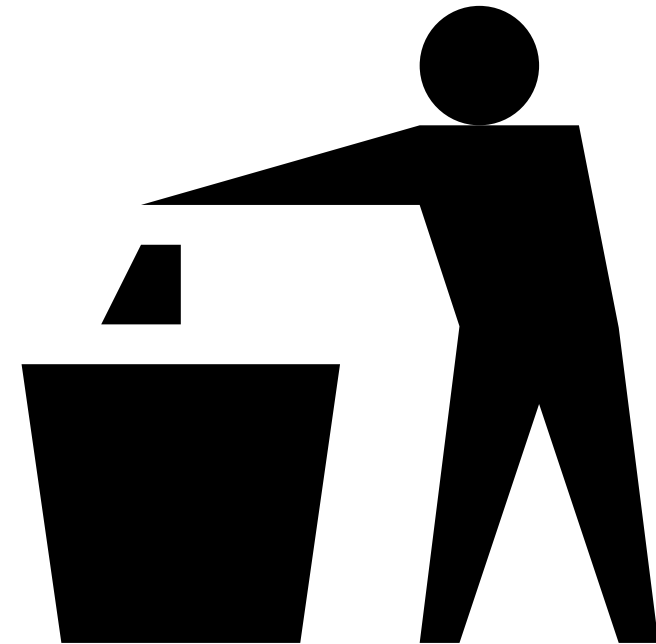
● Examples

- Electrocardiogram (ECG)
- Arterial Pressure Waveform (ABP)
- Photo-Plethysmogram (PPG)
- Electroencephalogram (EEG)



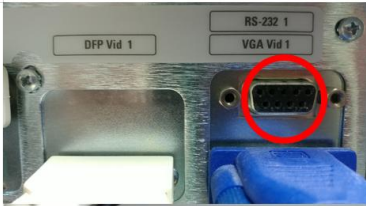
- Dynamic (계속 변한다) → Time-series
- Multi-factorial (다양한 인자) → High-dimensional (다채널)
- Interactive (의료인의 개입) → Imbalanced (대부분 정상)
- High-resolution (고해상도)
- Noisy
- Unstructured

Physiological Signal in real world practice



Vital Recorder

Solar8000



ROOT



Vigilance II



FMS



Drager



Vigileo



Hemosphere



PLEM



Datex-Ohmdea



EV100M



CardioQ



BIS



EV1000A



Orchestra



Invos



Vigilance



Agilia

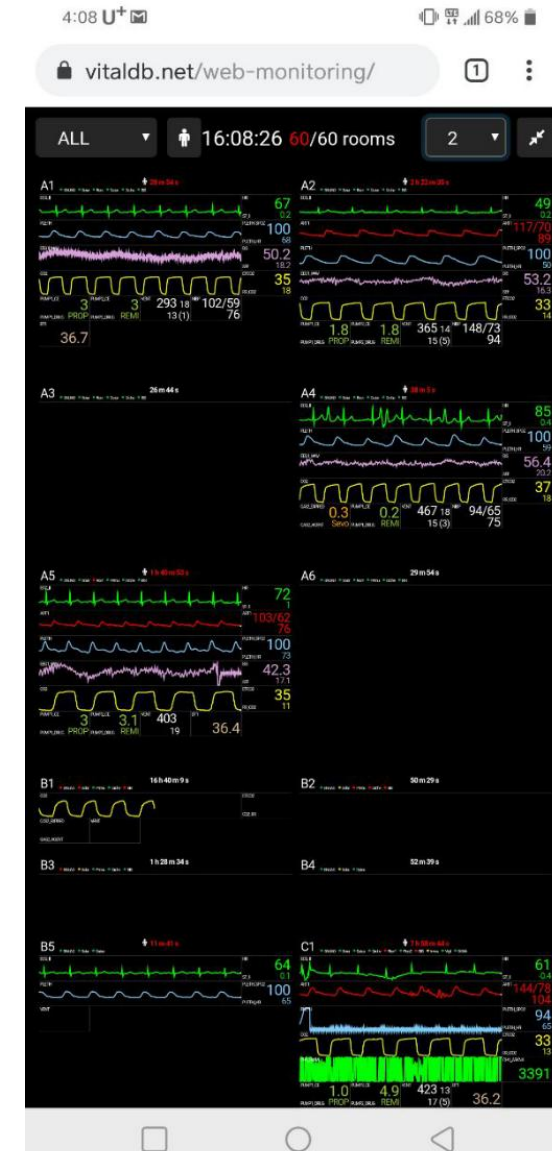
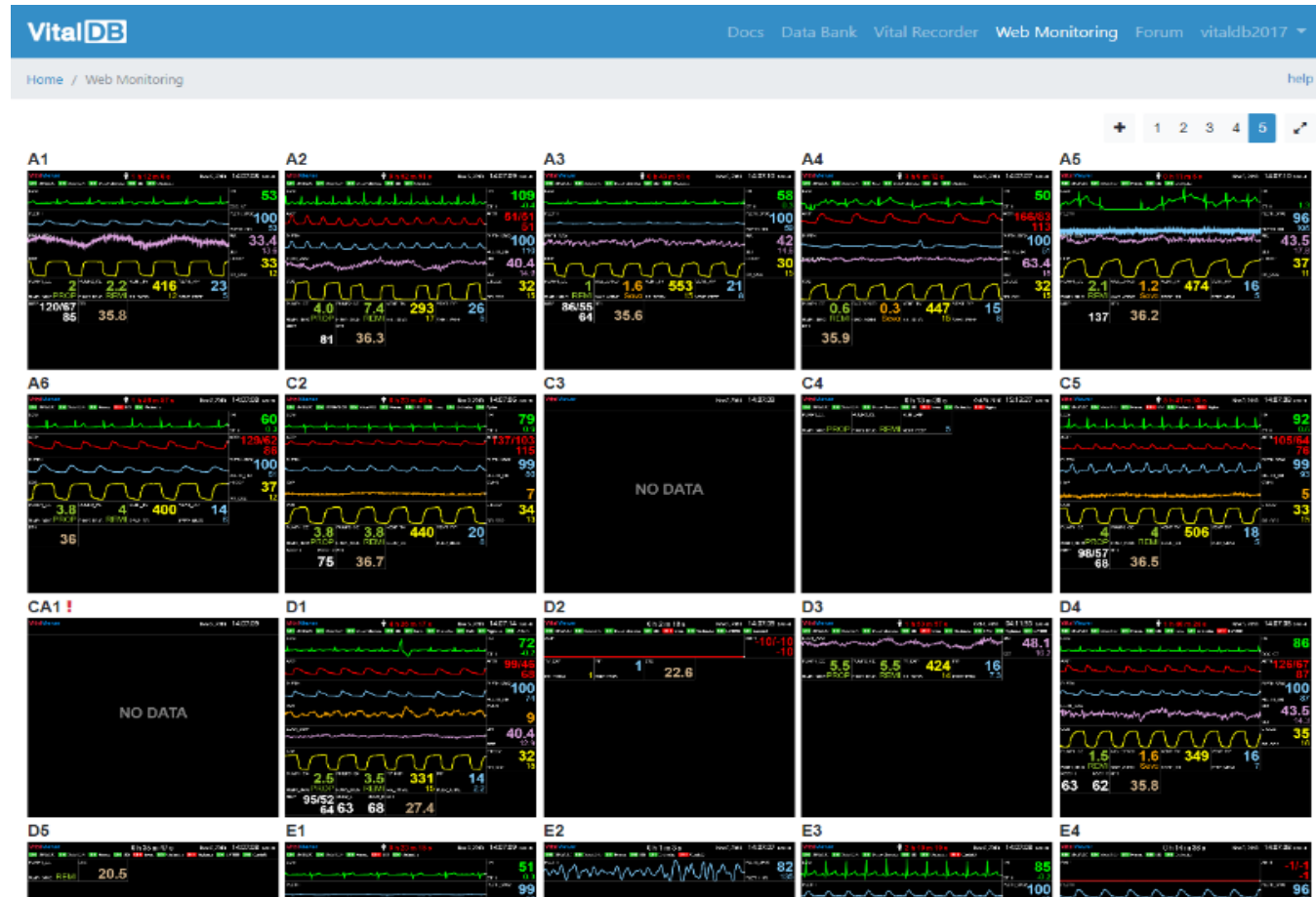


Vital Recorder



VitalDB Platform

Web Monitoring (since 2016)



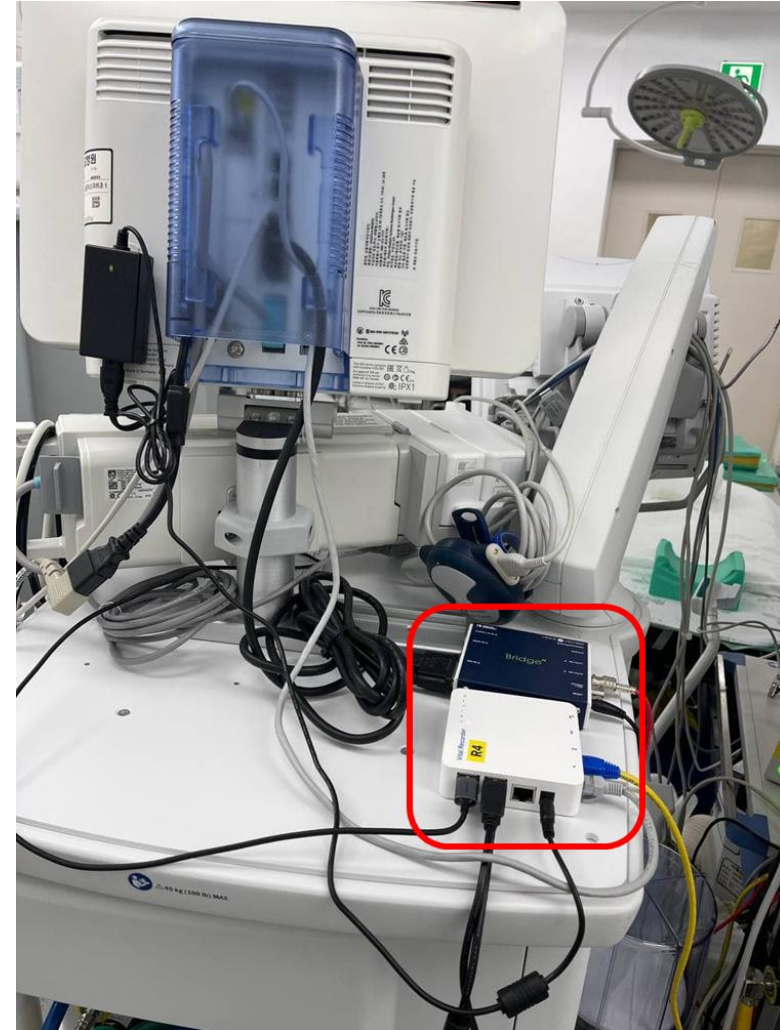
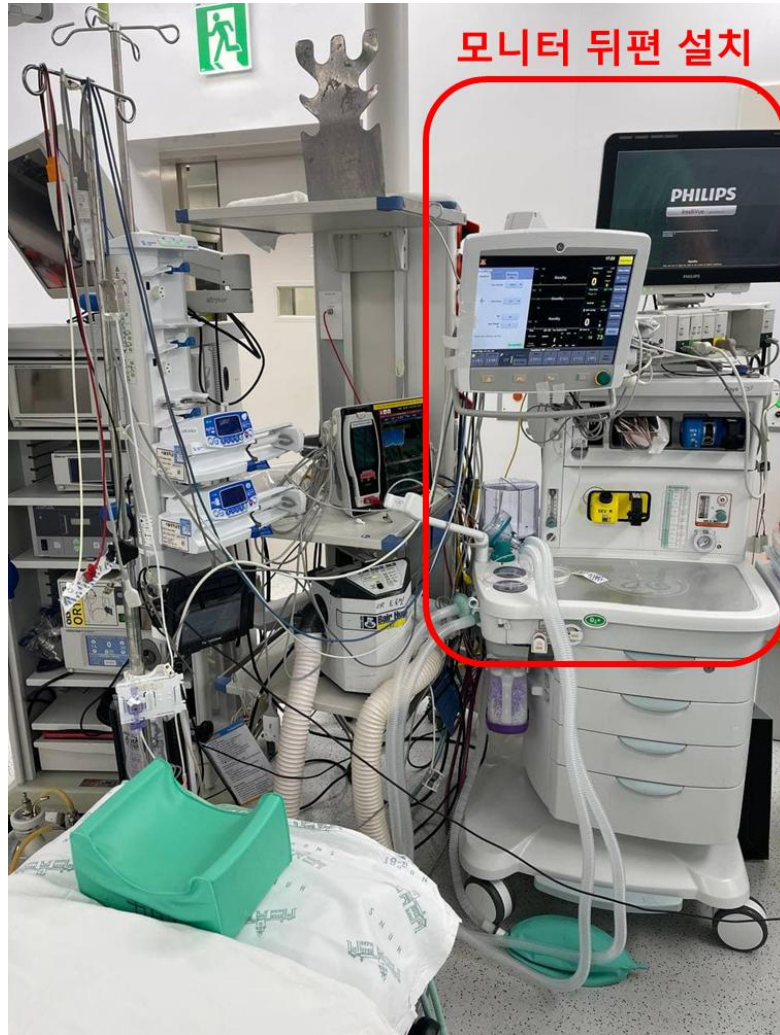
VitalDB Platform

- Clinical Usage: OR monitoring room



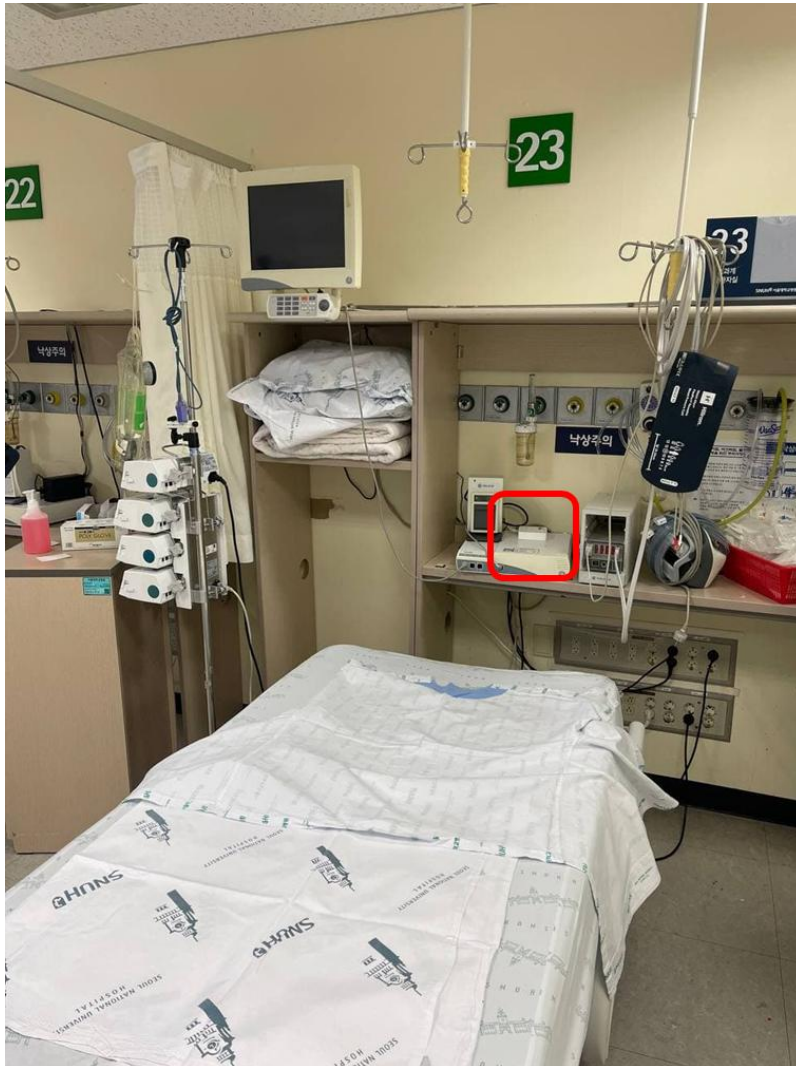
VitalDB Platform

- Clinical Usage: VR setup in OR



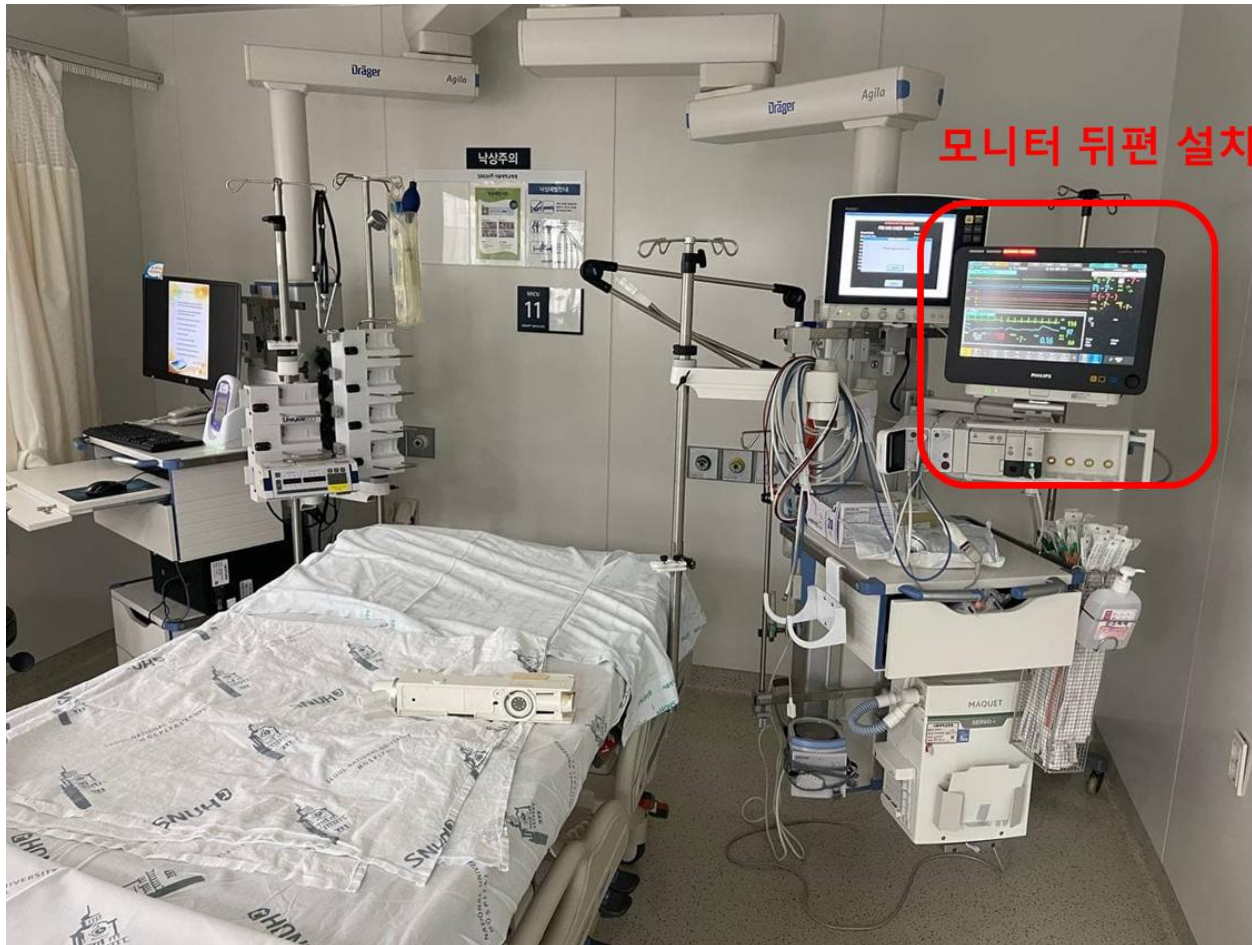
VitalDB Platform

- Clinical Usage: VR setup in SICU



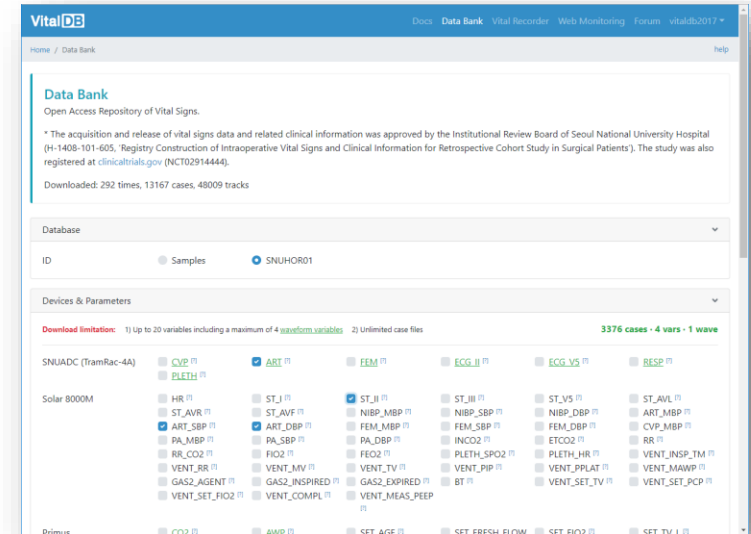
VitalDB Platform

- Clinical Usage: VR setup in MICU



VitalDB open dataset

- 세계 최대 수술환자 Open Dataset
 - 6,388 cases, 평균 2시간 반
 - 2016.8-2017.7 까지 1년간의 완전한 데이터 (GS, TS, UR)
- 수술 중 생체 신호 DB
 - 22억개의 수치값 (평균 80트랙)
 - 500Hz 심전도, Pleth (99%), 혈압파형 (55%)
 - 128Hz 뇌파 (87%)
 - Syringe Pump (55%)
 - Cardiac output (15%)



ID	Size (MB)	Age (years)	Sex	Height (cm)	Weight (kg)	ASA classification	Operation	Anesthesia type	Data duration (min)
00001	3.2	77	M	160.2	67.5	2	Low anterior resection	General	192
00004	5.7	74	M	160.6	53	2	Distal gastrectomy	General	350
00005	5.9	66	M	171	59.7	3	Aneurysmal repair	General	339
00007	4.3	52	F	167.7	62.3	2	Lung lobectomy	General	263
00010	5.7	72	M	162.5	62.75	3	Distal gastrectomy	General	350
00012	6.2	46	F	169.2	81.4	2	Liver transplantation	General	520
00013	2.9	67	F	153	64.9	2	Metastasectomy	General	180
00016	3.5	57	M	162.3	68.9	2	Liver segmentectomy	General	214
00017	5.6	85	M	164.2	53	2	Aneurysmal repair	General	339
00019	7.5	74	M	171.3	66.2	2	Pylorus preserving pancreaticoduodenectomy	General	460
00020	7.2	75	M	173.6	61.3	2	Lung lobectomy	General	441
00022	3.9	53	F	162	54.6	1	Total thyroidectomy	General	240
00024	1.8	57	M	161.8	65	2	Gastric wedge resection	General	108
00025	4	66	M	169.7	80.2	2	Total gastrectomy	General	247
00026	2.9	64	M	169.7	62.6	1	Lung lobectomy	General	178
00027	4.9	50	F	143.2	43.35	1	Distal gastrectomy	General	298
00028	7.3	67	F	148.9	49.9	1	Thymectomy	General	448
00029	5.8	53	F	156.1	57.4	1	Donor hepatectomy	General	337
00031	2.8	58	M	162.8	53.8	2	Lung wedge resection	General	174
00034	6.5	69	F	150.5	66.6	2	Breast-conserving surgery	General	396
00038	3.3	66	M	170.1	72.3	2	Lung lobectomy	General	205
00041	3.2	45	M	175	78.7	1	Nephrectomy	General	197

VitalDB open dataset

<https://vitaldb.net/dataset/>

Docs : VitalDB

vitaldb.net/docs/?documentId=1_VTteol5v-cdFkUTSnhYoGCqyq8dAC-_k50oM1QOjkU

VitalDB

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Introduction

Endpoints

Clinical informations

Track list

Track data

Format

Laboratory results

VitalDB Open Dataset API

Introduction

Anyone can access and use VitalDB dataset by programming. All API's can be called by GET method on the HTTPS standard protocol and downloaded by entering an address in the web browser.
The files have **GZip-compressed CSV** (comma separated values) format.
Each endpoint has the header at first row which contains column names.

Endpoints

Endpoint URL	Method	Description
https://api.vitaldb.net/cases	GET	Download clinical information
https://api.vitaldb.net/trks	GET	Download track list
https://api.vitaldb.net/tid	GET	Download track data {tid} : track identifier in track list
https://api.vitaldb.net/labs	GET	Download laboratory results

Clinical informations

Endpoint URL: <https://api.vitaldb.net/cases>
This endpoint contains the clinical information related to the case.

Columns	Data Source	Description	Unit
caseid	Random	Case ID; Random number between 00001 and 06388	
subjectid	EMR	Subject ID; Deidentified hospital ID of patient	
casestart	Case file	Recording Start Time; Set to 0 for anonymization	sec
caseend	Case file	Recording End Time; from casestart	sec
anestart	Case file	Anesthesia start time; from casestart	sec
aneend	Case file	Anesthesia end time; from casestart	sec

Dataset : VitalDB

vitaldb.net/dataset/?query=viewer

VitalDB

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Overview

API

Python Library

Data Viewer

AND 0/197

0/6388 1

Download CSV Fit width 100 Pixel/s

Search Tracks

Search Files (separate keywords with comma/newline, or set range with hyphen. e.g., 1-100,1145)

Track List

☐ SNUADC/ART

☐ SNUADC/CVP

☐ SNUADC/ECG_II

☐ SNUADC/ECG_V5

☐ SNUADC/FEM

☐ SNUADC/PLETH

☐ Solar8000/ART_DBP

☐ Solar8000/ART_MBP

☐ Solar8000/ART_SBP

☐ Solar8000/BT

☐ Solar8000/CVP

☐ Solar8000/ETCO2

☐ Solar8000/FEM_DBP

☐ Solar8000/FEM_MBF

☐ Solar8000/FEM_SBP

☐ Solar8000/FEO2

Custom select

Actions

CaseID	Len	Age	Sex	Operation
1	3h12m	77	M	Low anterior re...
2	4h22m	54	M	Subtotal gastre...
3	1h14m	62	M	Cholecystectomy
4	5h50m	74	M	Distal gastrect...
5	5h59m	66	M	Aneurysmal re...
6	1h26m	78	F	Cholecystectomy
7	4h22m	52	F	Lung lobectomy
8	1h39m	81	F	Breast-conservi...
9	1h14m	32	F	Cholecystectomy
10	5h49m	72	M	Distal gastrect...
11	1h28m	56	M	Transurethral r...
12	8h40m	46	F	Liver transplant...
13	3h00m	67	F	Metastasectomy
14	1h08m	23	M	Cholecystectomy
15	0h59m	56	F	Hernia repair...

SNUADC

ART

ECG_II

ECG_V5

PLETH

Solar8000

ART_DBP

ART_MBP

ART_SBP

ETCO2

FEO2

FIO2

GAS2_AGENT

GAS2_EXPIRED

GAS2_INSPiRED

HR

INCO2

NIBP_DBP

NIBP_MBP

NIBP_SBP

PLETH_HR

PLETH_SPO2

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VitalDB open dataset

- **Demographic**
 - Age; Sex; Height; Weight; BMI; ASA classification; Emergency;
- **Surgery & Anesthesia**
 - Department; Surgery type; Diagnosis; Operation; Surgical approach; Surgical position; Anesthesia type; Sevoflurane; Desflurane; Propofol TCI; Remifentanyl TCI; Surgery duration; Anesthesia duration
- **Preoperative data**
 - Hypertension; Diabetes; Arrhythmia; Pulmonary function; Hemoglobin; Albumin; Cr; Na⁺; K⁺
- **Intraoperative data**
 - Cormack's grade; Airway route; Endotracheal tube size; Double lumen tube size; LMA size; IV line; Arterial line; Central line; Estimated blood loss; Urine output; RBC; FFP; Crystalloid; Colloid; Propofol bolus; Midazolam bolus; Fentanyl bolus; Rocuronium; Vecuronium; Ephedrine; Phenylephrine; Epinephrine; Calcium chloride
- **Outcomes**
 - Postoperative stay; Postoperative ICU stay; Mortality

Clinical Information

caseid	casestart	caseend	adm	dis	anestart	aneend	opstart	opend	age	sex	los	height	weight	los_posto	los_icu	death_inh
1	0	11520	-236220	627780	-540	10860	-540	10380	77	M	10	160.2	65.5	8	0	0
2	0	15720	-221160	1506840	-1020	14940	1740	14640	54	M	20	167.3	56	18	0	0
3	0	4440	-218640	40560	-540	4260	-540	3960	62	M	3	169.1	69.7	1	0	0
4	0	21000	-201120	576480	-720	20280	2580	18180	74	M	9	160.6	53.7	7	0	0
5	0	21540	-67560	3734040	-960	22440	2640	21840	66	M	44	171	69.5	43	10	0
6	0	5160	-300600	131400	0	4800	2700	4200	78	F	5	150	54.6	2	0	0
7	0	15720	-154320	623280	480	14820	480	14280	52	F	9	167.5	63.8	8	3	0
8	0	5940	-219720	212280	-720	5280	960	4980	81	F	5	156.7	67.25	3	0	0
9	0	4440	-135900	36900	-480	4500	-480	3900	32	F	2	157.9	50.9	1	0	0
10	0	20940	-220740	3580860	-1740	21060	2460	20460	72	M	44	162	66.7	42	0	0
11	0	5280	-135540	37260	-180	5160	660	4560	56	M	2	175.4	81.45	1	0	0
12	0	31200	-208500	1519500	-180	31500	-180	30900	46	F	20	169.2	88.2	18	4	0
13	0	10800	-114660	230940	-660	10740	-660	9540	67	F	4	152.5	61.25	3	0	0
14	0	4080	-206640	52560	-240	4260	1260	3360	23	M	3	177.9	80	1	0	0
15	0	3540	-150360	22440	-900	3240	-900	2640	56	F	2	158	48.25	1	0	0
16	0	12840	-136920	381480	-1020	13680	2580	11880	57	M	6	162.3	69.15	5	0	0
17	0	20340	-380160	1002240	-180	20340	3540	19740	85	M	16	164.2	53.7	12	0	0
18	0	4440	-407580	197220	-780	4020	-780	3420	77	F	7	155	57.35	3	0	0
19	0	27600	-211380	1343820	-180	27720	2220	27120	74	M	18	171.3	65.4	16	0	0
20	0	26460	-822780	1164420	120	27120	3420	26220	75	M	23	173.6	60.1	14	1	0
21	0	12660	-311640	465960	-360	12660	-360	12060	69	F	9	155.1	62.55	6	0	0
22	0	14400	-114660	230940	-660	14040	2040	13440	53	F	4	162	54.6	3	0	0
23	0	2820	-116640	142560	-2640	2760	-2640	2160	69	F	3	139.4	47.8	2	0	0

Track information

caseid	ttype	tname	unit	srate	gain	bias	downloadurl			
1	N	Primus/RR_VF	/min	0	1	0	https://vitaldb.net/trk.php?tid=07efbdc02ef32a73			
1	N	Solar8000/PLETH_HR		0	1	0	https://vitaldb.net/trk.php?tid=07f5af76886e2fd0			
1	N	Solar8000/ART_MBP	mmHg	0	1	0	https://vitaldb.net/trk.php?tid=0aea1434883377c9			
1	N	Solar8000/GAS2_INSPIRED	%	0	1	0	https://vitaldb.net/trk.php?tid=0dab8e62fe9327ef			
1	N	Primus/EXP_DES	kPa	0	1	0	https://vitaldb.net/trk.php?tid=0ebfe07eb4f71b56			
1	W	SNUADC/ART	mmHg	500	0.987457	-496.787	https://vitaldb.net/trk.php?tid=0f4441821bbf4b67			
1	N	Primus/INCO2_KPA	kPa	0	1	0	https://vitaldb.net/trk.php?tid=121a0106bbc3092a			
1	N	Solar8000/ST_I	mm	0	1	0	https://vitaldb.net/trk.php?tid=1a67d07bc84a6113			
1	N	Solar8000/VENT_TV	mL	0	1	0	https://vitaldb.net/trk.php?tid=1b184d5317c08d02			
1	W	Primus/CO2	mmHg	62.5	0.075	0	https://vitaldb.net/trk.php?tid=1b460fd0fbdf3459			
1	N	Primus/ETCO2_KPA	kPa	0	1	0	https://vitaldb.net/trk.php?tid=1c487ef269e3479b			
1	W	BIS/EEG1_WAV	uV	128	1	0	https://vitaldb.net/trk.php?tid=1c9ad0d4af60fb39			
1	N	Solar8000/ART_DBP	mmHg	0	1	0	https://vitaldb.net/trk.php?tid=229e9b60bf33549e			
1	N	BIS/BIS		0	1	0	https://vitaldb.net/trk.php?tid=260460a8927afa06			
1	N	Primus/EXP_SEVO	kPa	0	1	0	https://vitaldb.net/trk.php?tid=2a1df32a875cf033			
1	W	BIS/EEG2_WAV	uV	128	1	0	https://vitaldb.net/trk.php?tid=2d1217c48aac3737			
1	N	Solar8000/ST_V5	mm	0	1	0	https://vitaldb.net/trk.php?tid=323138ef9a2b1509			
1	N	Primus/ETCO2	mmHg	0	1	0	https://vitaldb.net/trk.php?tid=3679e6c9d7f05a11			
1	N	Solar8000/FIO2	%	0	1	0	https://vitaldb.net/trk.php?tid=39a2a6558926873a			
1	N	Primus/SET_INSP_PAUSE	%	0	1	0	https://vitaldb.net/trk.php?tid=3b062ef08d95750d			

Track types

Wave Track

ADC Counts

605
604
604
605
606
606
606
607
608
610
611
610
611

Numeric Track

Time Value

490.247	3.3
492.247	3.2
494.743	2.7
496.246	3.2
498.727	3.6
500.247	3.8
502.246	3.6
504.726	3.3
506.726	3.5
509.43	3.4
513.845	3.5
516.245	3.5
518.725	3.4

String Track

Time Value

363.845	Sevoflurane
365.848	Sevoflurane
367.846	Sevoflurane
369.846	Sevoflurane
371.844	Sevoflurane
373.845	Sevoflurane
375.845	Sevoflurane
377.844	Sevoflurane
379.843	Sevoflurane
381.842	Sevoflurane
383.842	Sevoflurane
385.842	Sevoflurane
387.841	Sevoflurane
389.857	Sevoflurane

Physically Measured Value = ADC count * gain + bias

Time = seconds from case start
2100-01-01 00:00:00 GMT

Low blood pressure prediction

● Goal

- 20초간의 arterial wave로부터 1분 뒤 blood pressure 60 이하를 예측할 수 있는 딥러닝 모델 개발
- 수술 중 기록된 arterial waveform 에서 추가적인 intervention 없이 low blood pressure 발생을 예측할 수 있는 CNN 기반 모델
- Min-max 필터 적용 (상한 하한 적용)
- Mean standard deviation 필터 적용

● Track information

- Arterial waveform: 'SNUADC/ART'
- Mean blood pressure: 'Solar8000/ART_MBP'

● Environmental setup

- Preprocessor 모듈을 불러오기 위해 아래와 같이 preprocessor.py 경로를 지정해주어야 함
- 다운로드 받은 폴더의 경로를 지정
- (실습 .ipynb 파일과 동일 경로에 있다면 바로 로드 가능)

```
import sys
sys.path.append('/This/is/my/download/directory/')
```

Low blood pressure prediction

● Data loading

```
chart_pd_01 = pd.read_csv('{}/sample_csv/{}'.format(download_directory, '00010.csv'))

selected_tracks = ["SNUADC/ART", "Solar8000/ART_MBP"]

chart_pd_01_trks = chart_pd_01[selected_tracks]
print(chart_pd_01_trks.columns)
```

● mean blood pressure 데이터 추출

```
# mbp 데이터 추출
mbp_data_pd = chart_pd_01[col_mbp][chart_pd_01[col_mbp].notnull()]
mbp_index = mbp_data_pd.index.values
print(mbp_data_pd)
```

● arterial wave 데이터 추출 및 nan 값 지우기

```
# art 데이터 전체 추출 및 nan value 채우기
art_full_pd = chart_pd_01[col_art]
art_full_pd = art_full_pd.fillna(0)
print(art_full_pd)
```

Low blood pressure prediction

arterial segment 추출

```
# 실습: arterial segment 추출하기
mbp_values_list = [mbp_data_pd[idx] for idx in mbp_points]
art_seg_list = [art_full_pd[idx-(mdelay*60*srate)-(length*srate):idx-(mdelay*60*srate)].values for idx in mbp_points]
```

조건에 따라 filter들 정의

```
# 필터들 선언
# mbp min-max filter
mbp_max_filter = mbp_values_np < max_limit_mbp
mbp_min_filter = mbp_values_np > min_limit_mbp
mbp_filter = mbp_max_filter & mbp_min_filter

# abp range filter
art_filter_list = []
for seg in art_seg_np:
    filter_value = (np.array(seg) > 25.0).all() and (np.array(seg) < 250.0).all()
    art_filter_list.append(filter_value)
art_filter = np.array(art_filter_list)

# mstds 필터
mstds_values_list = []
for seg in tqdm(art_seg_np):
    if (np.array(seg) < 0.).any():
        mstds_values_list.append(float(0.))
    else:
        mstd_val, _ = pre.process_beat(seg)
        mstds_values_list.append(mstd_val)
mstds_filter = np.array(mstds_values_list) > 0.
```

100% |  | 10432/10432 [02:15<00:00, 77.27it/s]

Low blood pressure prediction

● 필터 적용하여 데이터 추출

```
# 필터 적용해서 추출
mbp_filtered = mbp_values_np[all_filters]
art_filtered = art_seg_np[all_filters]
print(mbp_filtered.shape)
print(art_filtered.shape)
```

● binary label 정의

```
label_tmp = np.zeros(len(mbp_filtered))
label_tmp[mbp_filtered<61] = 1.
```


Low blood pressure prediction

● 사전에 구축된 데이터셋 로드

```
with open('{} /datasets/lowbp_alines_1min_delay_20sec_compressed.np'.format(download_directory), 'rb') as f:
    aline_data = pickle.load(f)

with open('{} /datasets/lowbp_ylabels_1min_delay_20sec_compressed.np'.format(download_directory), 'rb') as f:
    ylabel_data = pickle.load(f)

with open('{} /datasets/lowbp_charts_1min_delay_20sec_compressed.np'.format(download_directory), 'rb') as f:
    charts_data = pickle.load(f)

with open('{} /datasets/lowbp_mbps_1min_delay_20sec_compressed.np'.format(download_directory), 'rb') as f:
    mbps_data = pickle.load(f)
```

● train / validation set 구분

```
valid_index = (charts_data > '00085.csv')
train_index = ~valid_index

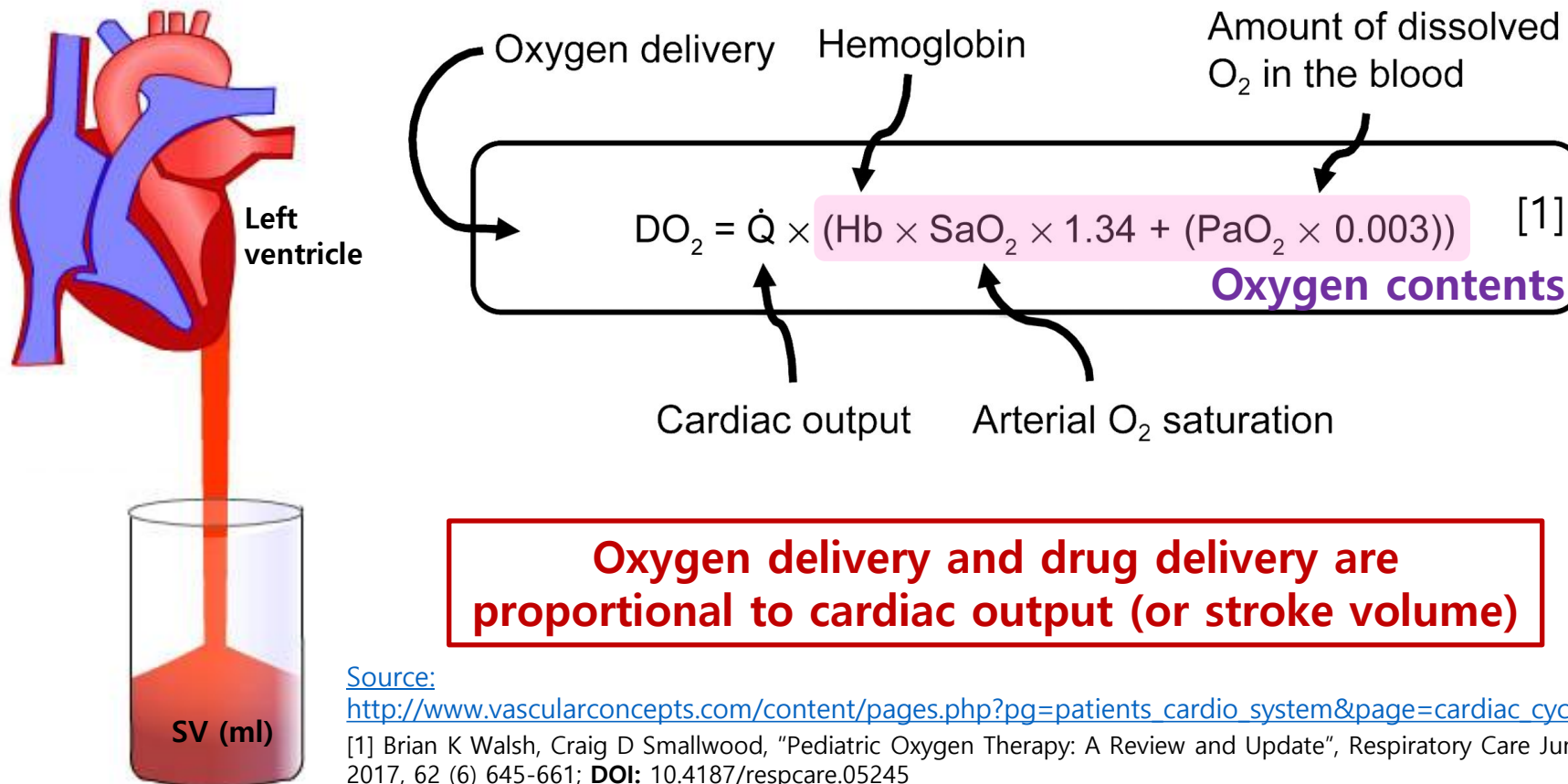
aline_train = aline_data[train_index]
ylabel_train = ylabel_data[train_index]

aline_valid = aline_data[valid_index]
ylabel_valid = ylabel_data[valid_index]
```

Stroke volume prediction

● Cardiac Output (Stroke Volume) estimation

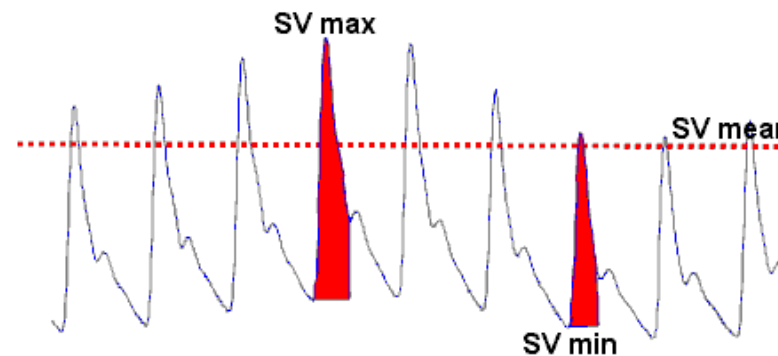
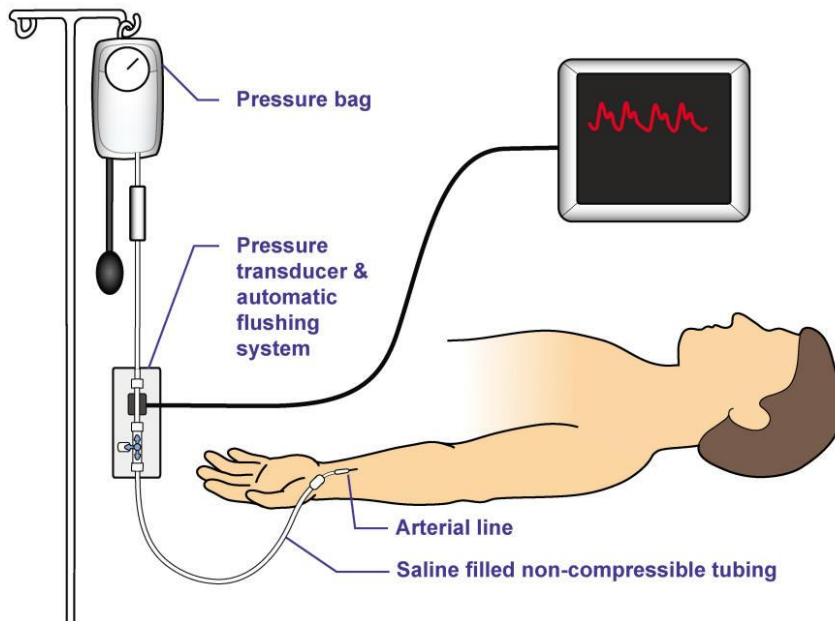
- Stroke Volume (SV): Volume of blood pumped from a left ventricle per beat
- Cardiac Output (CO) = SV × Heart Rate (HR)



Stroke volume prediction

● Arterial Pressure based Cardiac Output (APCO)

- It measures the amount of blood moving when a stroke occurs from arterial wave
- Arterial line is used for almost every patients in operations to real-time monitoring of blood pressure
- **Less invasive than Pulmonary Artery Catheter based Cardiac Output (PAC-CO)**
- EV1000 (Edward Lifescience), Vigileo (Edwards Lifescience), ...



Stroke volume prediction

● Goal

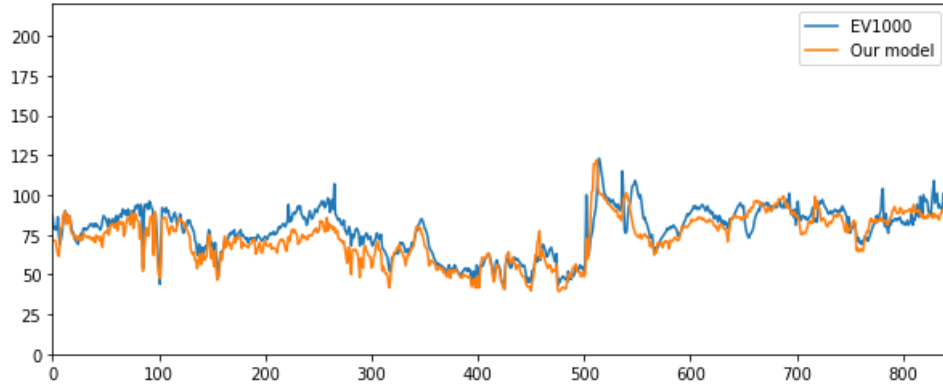
- 20초간의 arterial wave로부터 Stroke Volume을 예측할 수 있는 딥러닝 모델 개발
- 수술 중 기록된 arterial waveform 에서 추가적인 intervention 없이 stroke volume을 계산할 수 있는 CNN 기반 Regression모델
- Min-max 필터 적용 (상한 하한 적용)
- Mean standard deviation 필터 적용

● Track information

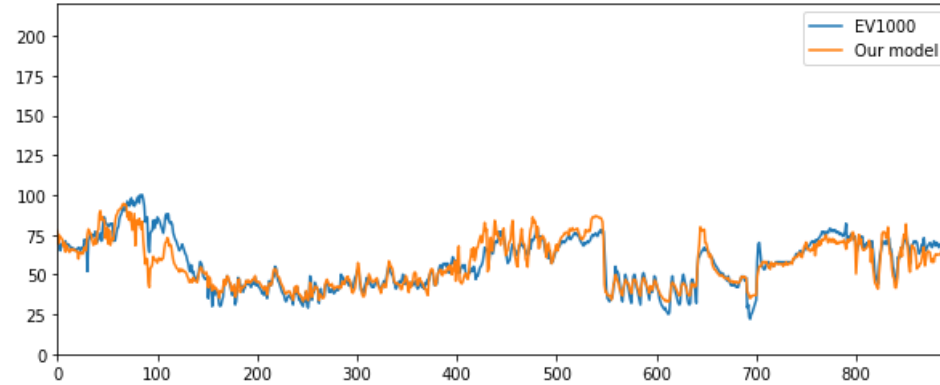
- Arterial waveform: 'SNUADC/ART'
- Stroke volume: 'EV1000/SV'

Stroke volume prediction

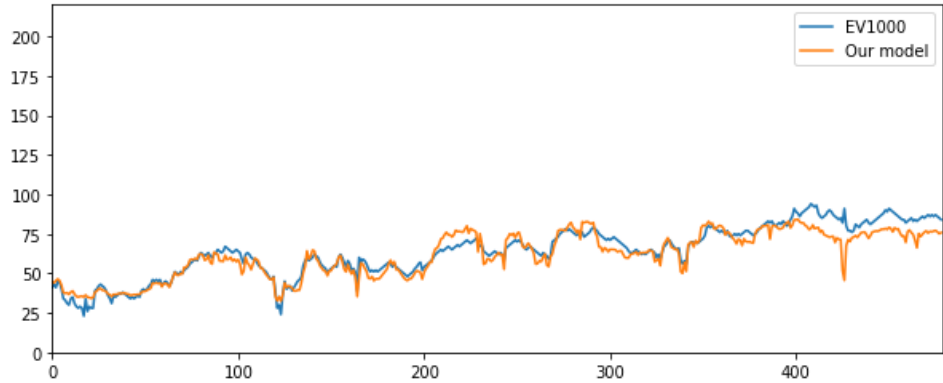
CaseID : 00183



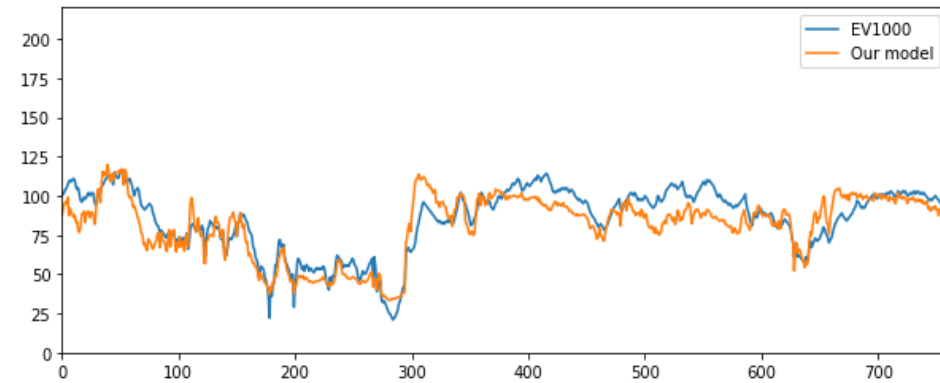
CaseID : 00185



CaseID : 00187

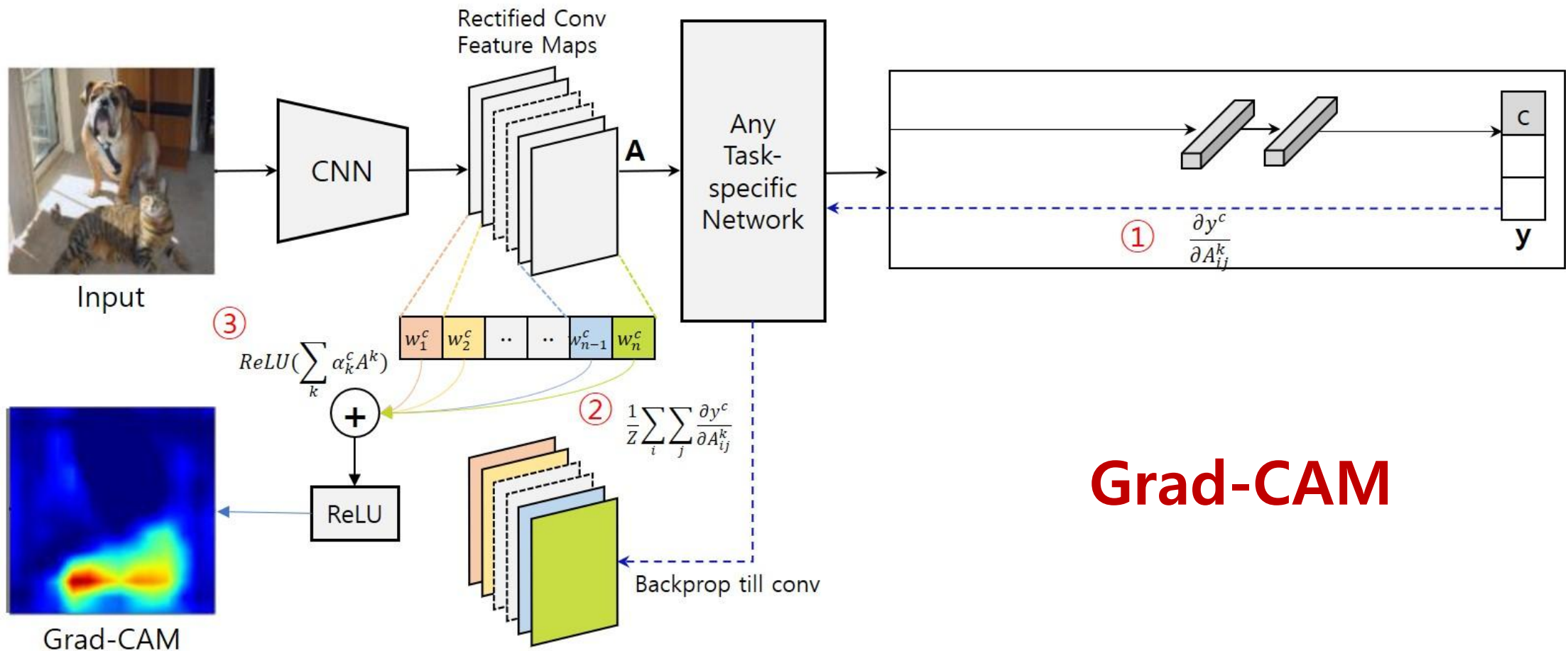


CaseID : 00189



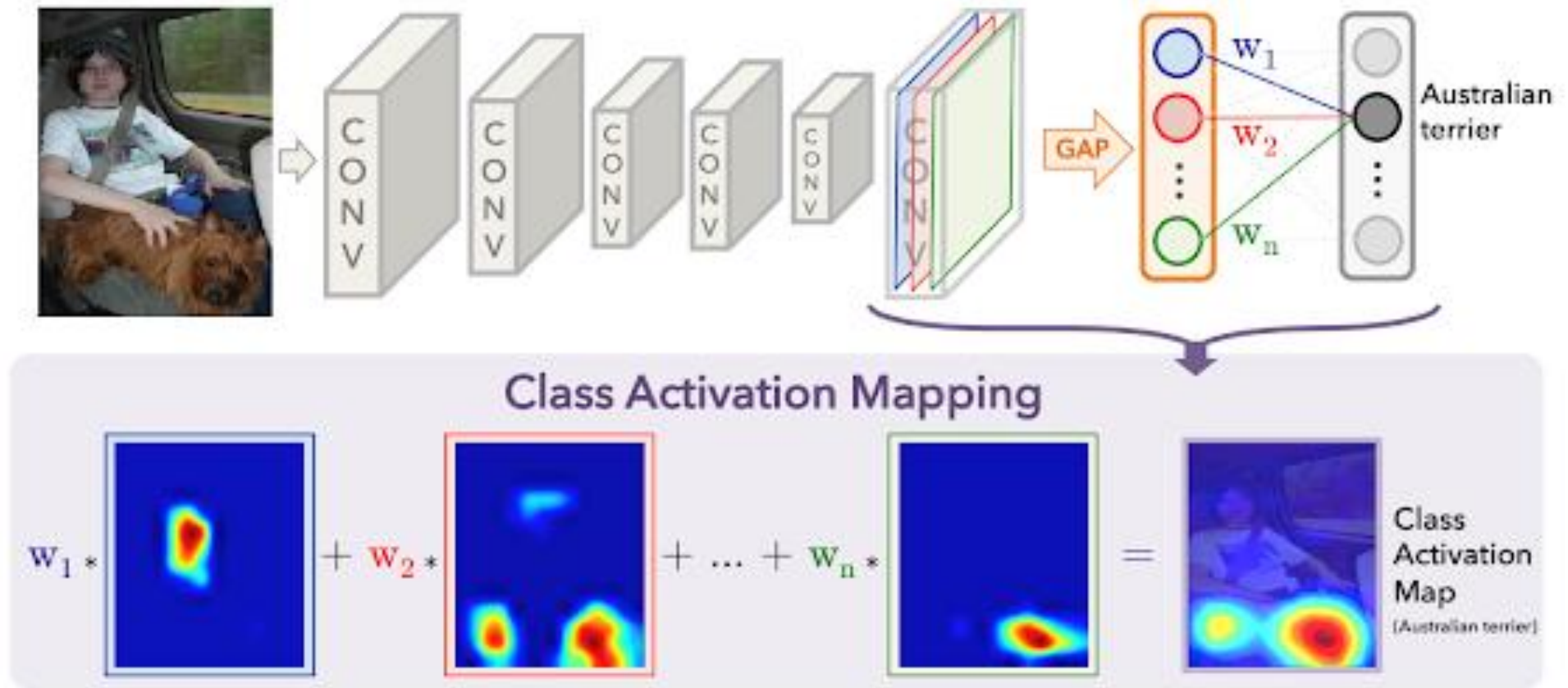
Post-hoc XAI method 1: Grad-CAM

- Post-hoc: 모델이 학습된 이후에 적용하는 방법
- eXplainable AI (XAI): 설명가능한 인공지능 방법론



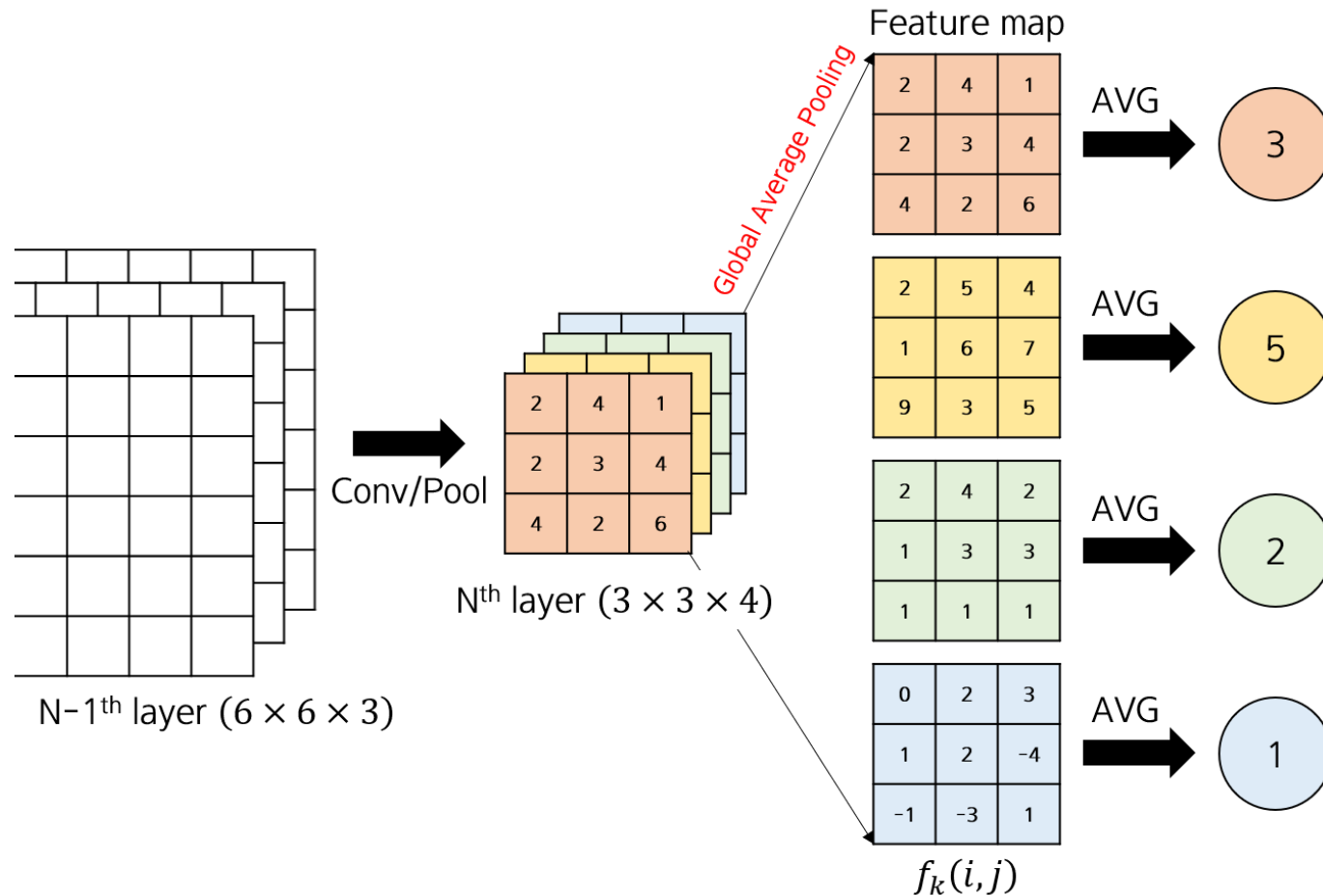
Post-hoc XAI method 1: Grad-CAM

- Initial Proposal: CAM (Class Activation Map)



Post-hoc XAI method 1: Grad-CAM

Global Average Pooling (CAM)

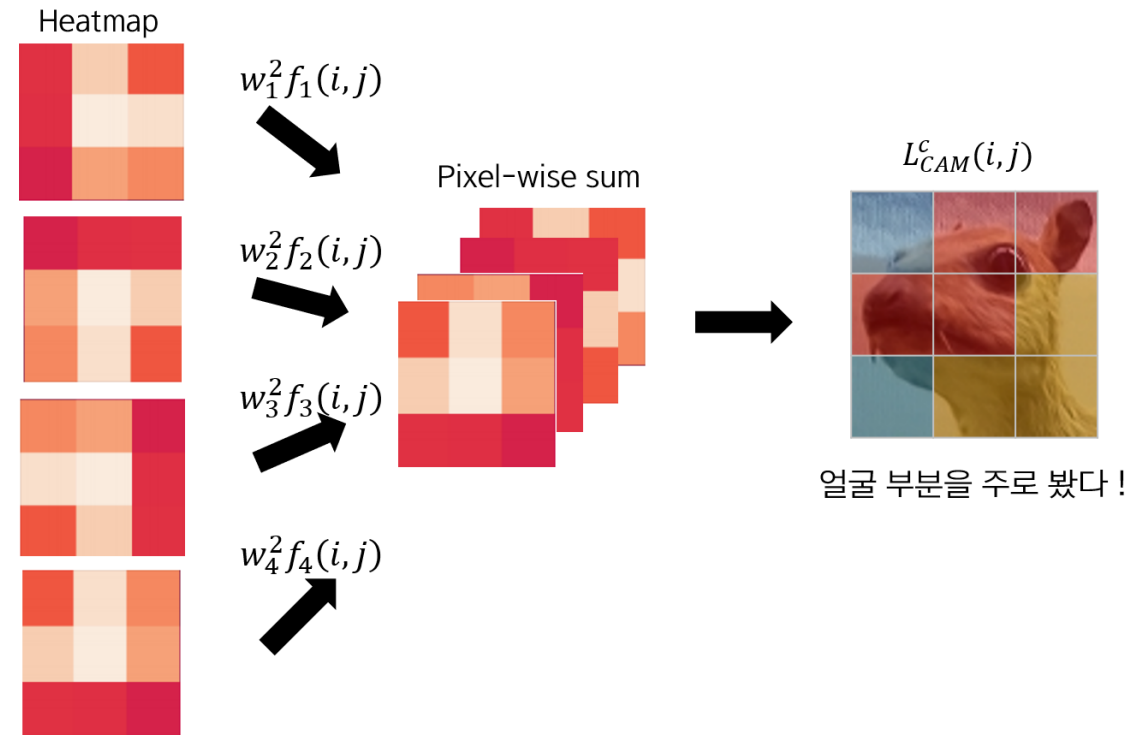
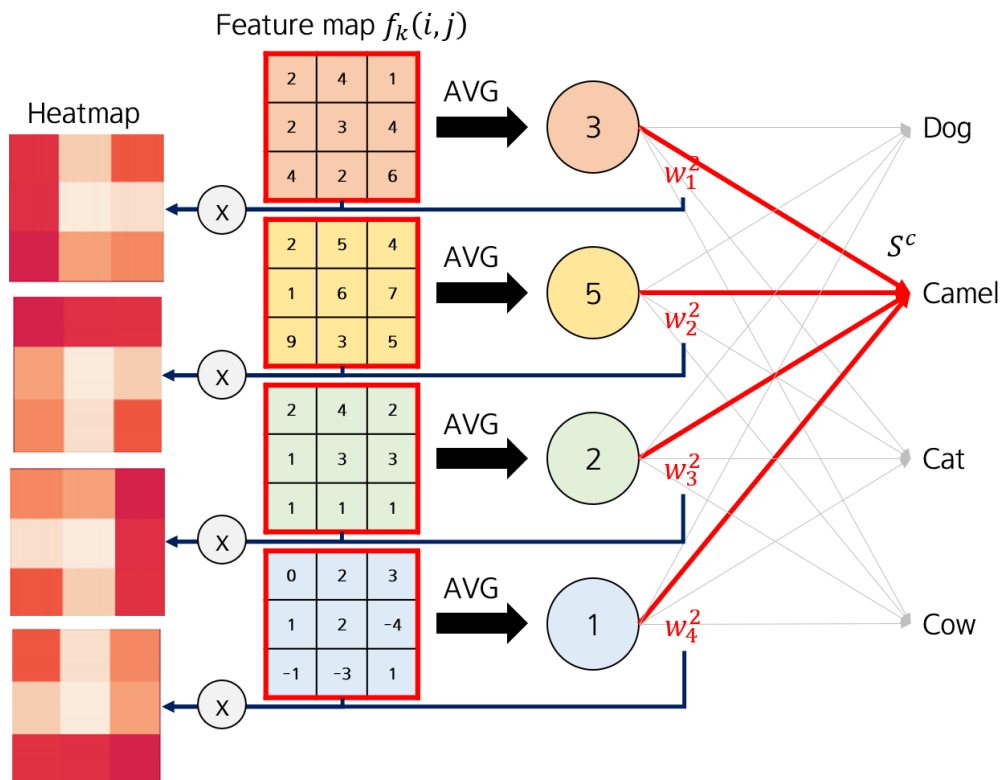


Post-hoc XAI method 1: Grad-CAM

Class activation mapping (CAM)

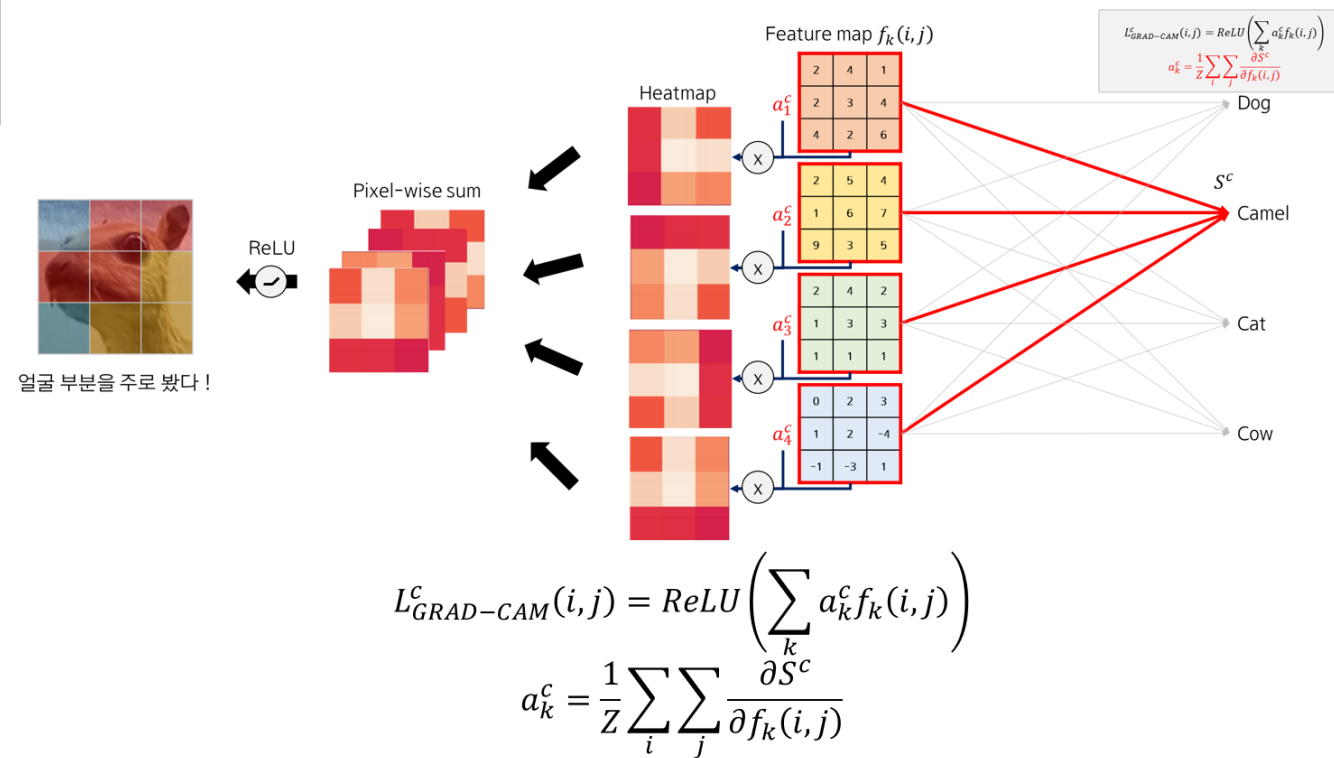
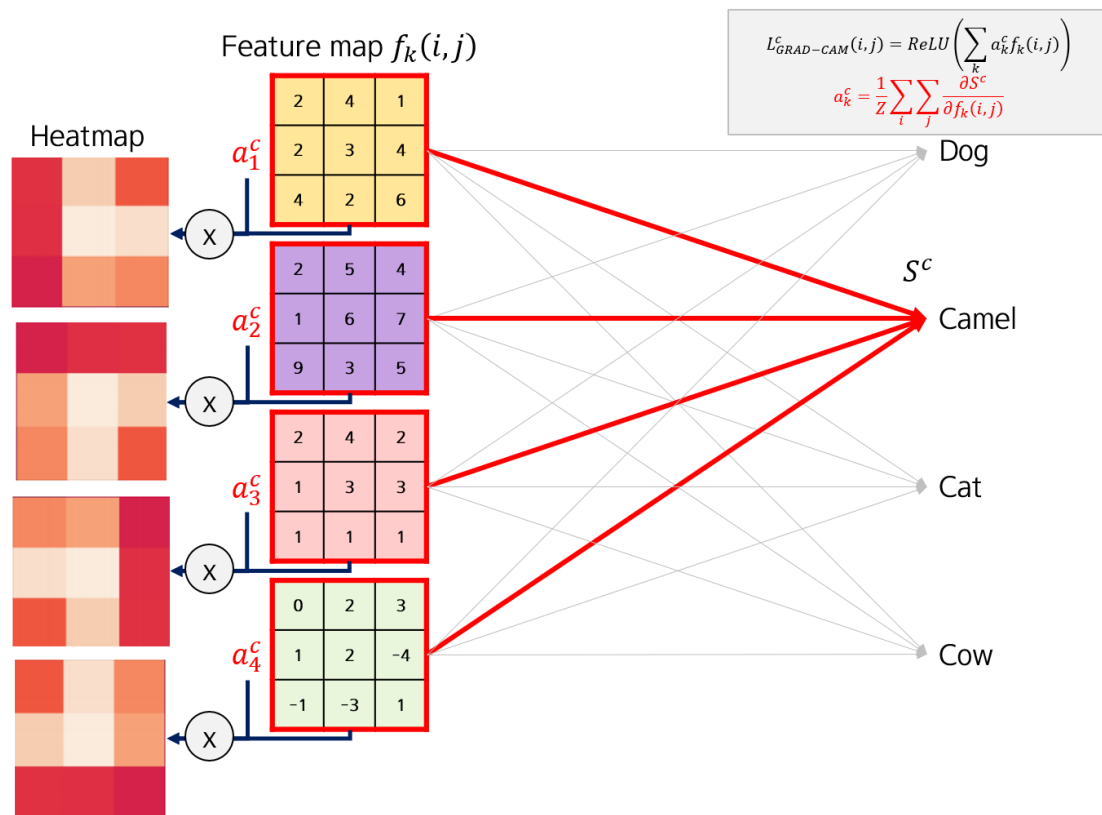
Limitation 1: GAP가 반드시 필요함

Limitation 2: 마지막 conv layer 에 대해서만 추출 가능



Post-hoc XAI method 1: Grad-CAM

- Gradient-weight CAM (**Grad-CAM**) Solution: feature weight를 gradient 로 대체하자!



Post-hoc XAI method 1: Grad-CAM

Grad-CAM for "Cat"



Grad-CAM for "Dog"



Is it an appropriate method for physiological signal processing?

Post-hoc XAI method 2: Integrated Gradient

- Approach: Gradient-based XAI

Input Image



Predicted Label:
Persian cat



Predicted Label:
Ringlet butterfly



Predicted Label:
Soap dispenser



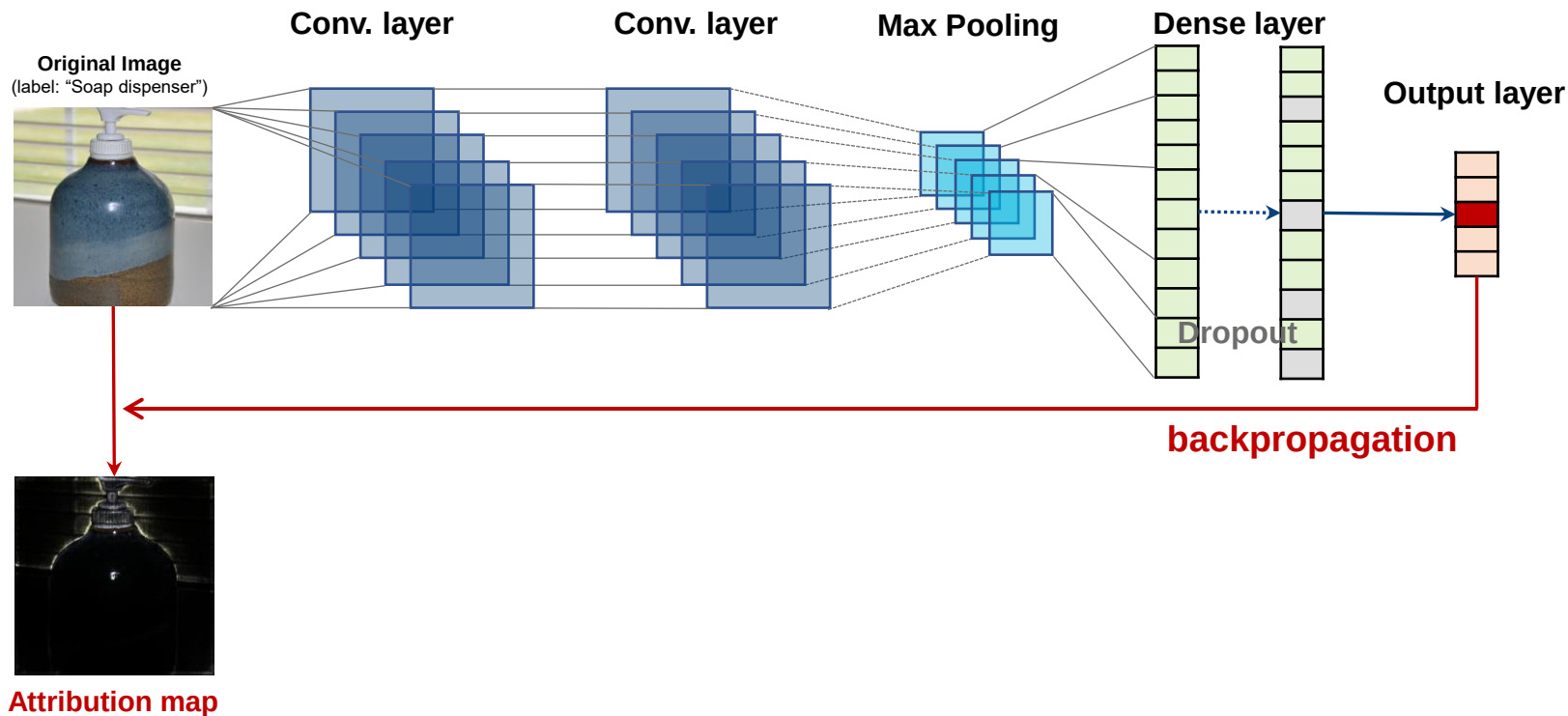
Attribution map



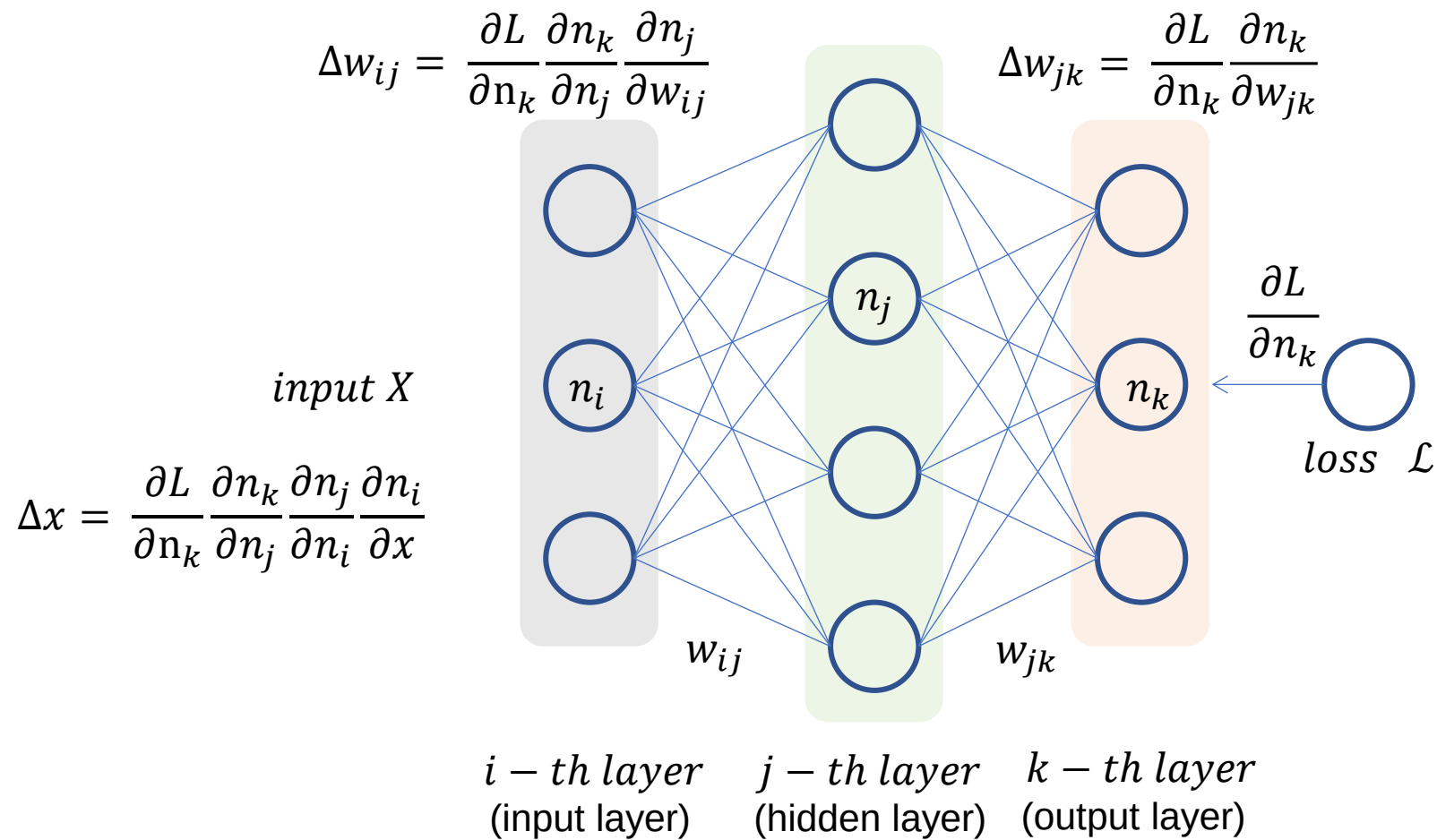
Post-hoc XAI method 2: Integrated Gradient

Advantages of Gradient-based methods

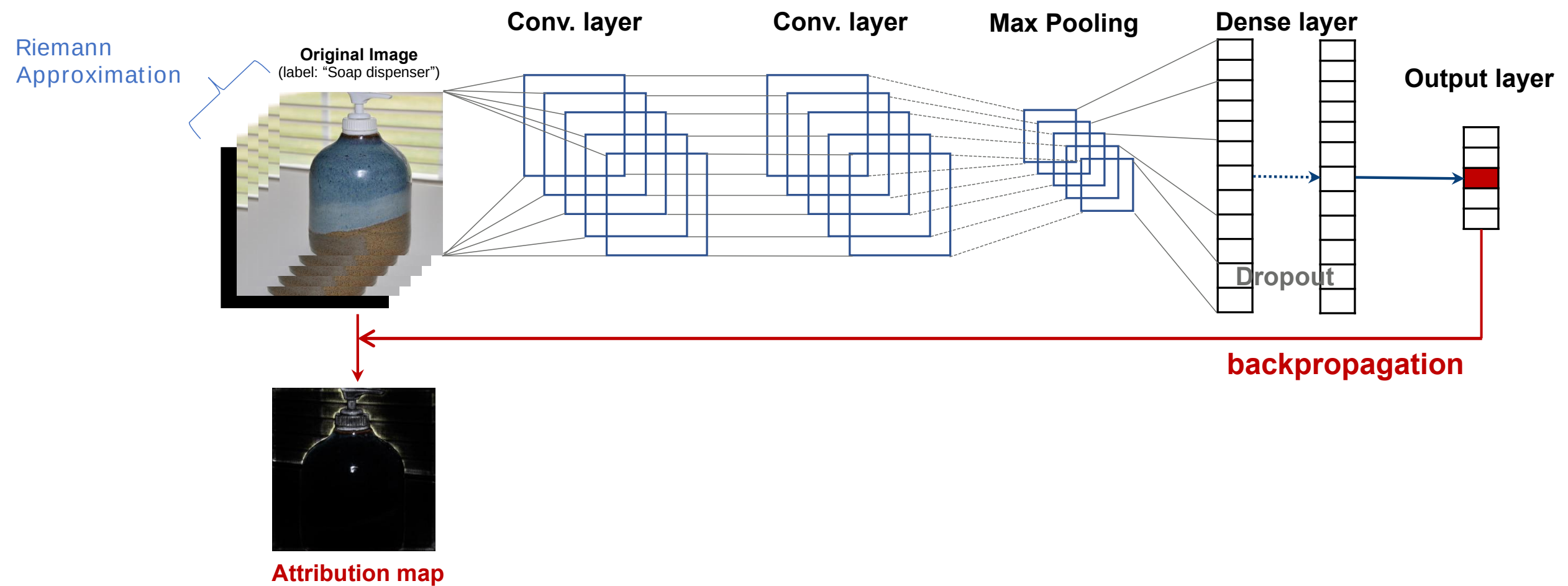
- Unnecessary additional models except the trained model itself
- It is possible to end-to-end learning
- Easy to compute and post-hoc method



Post-hoc XAI method 2: Integrated Gradient

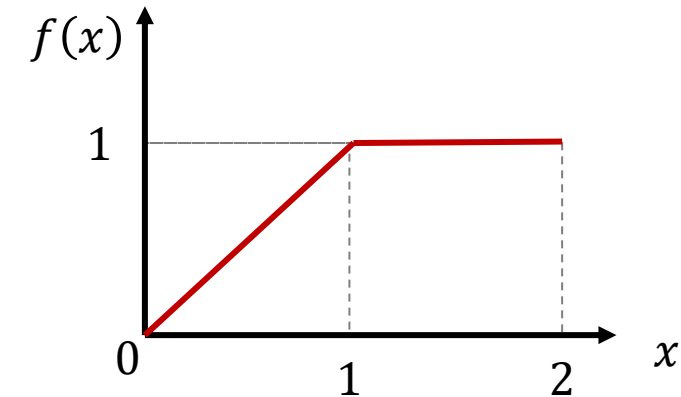
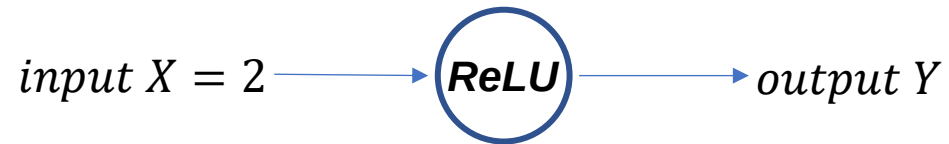


Post-hoc XAI method 2: Integrated Gradient

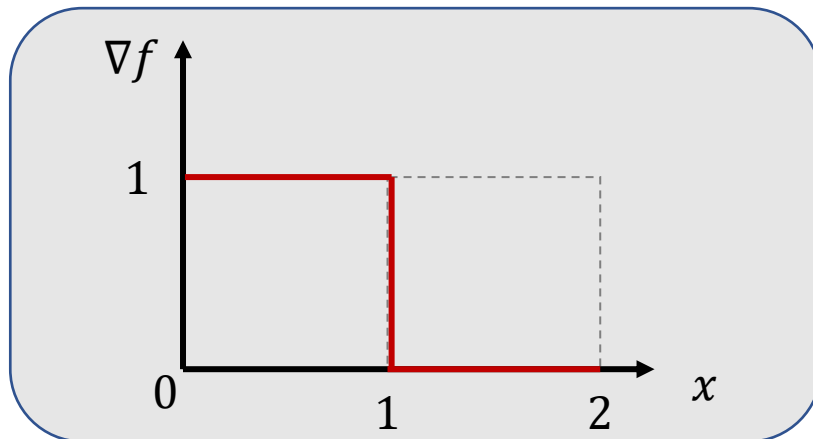


Post-hoc XAI method 2: Integrated Gradient

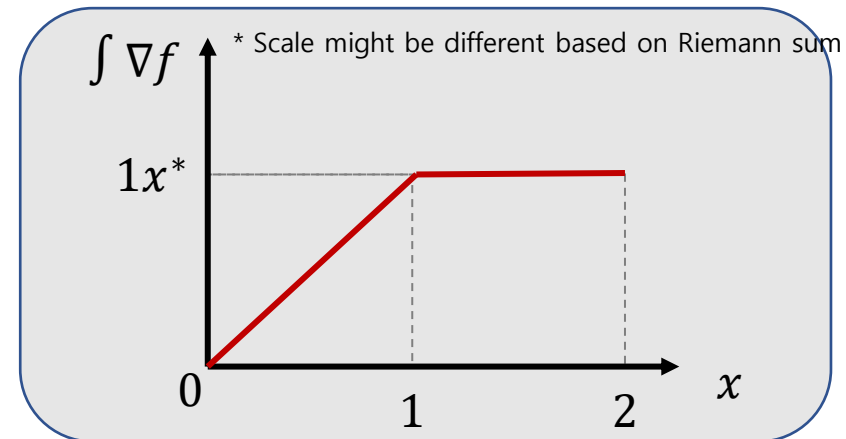
$$f(x) = 1 - \text{ReLU}(1 - x)$$



Gradient respect to x



Cumulating gradient respect to x



Post-hoc XAI method 2: Integrated Gradient



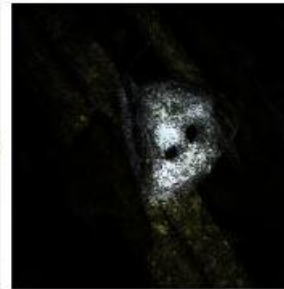
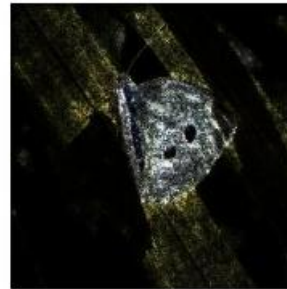
Top label: mosque
Score: 0.999127



Top label: viaduct
Score: 0.999994



Top label: cabbage butterfly
Score: 0.996838

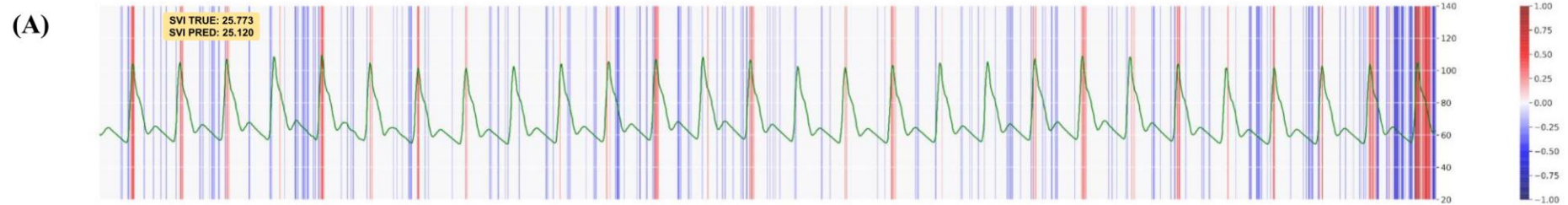


Top label: starfish
Score: 0.999992

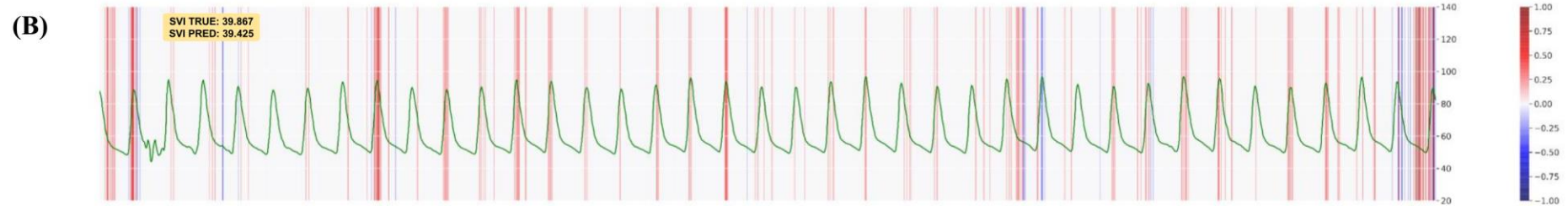


Interpretable AI example results

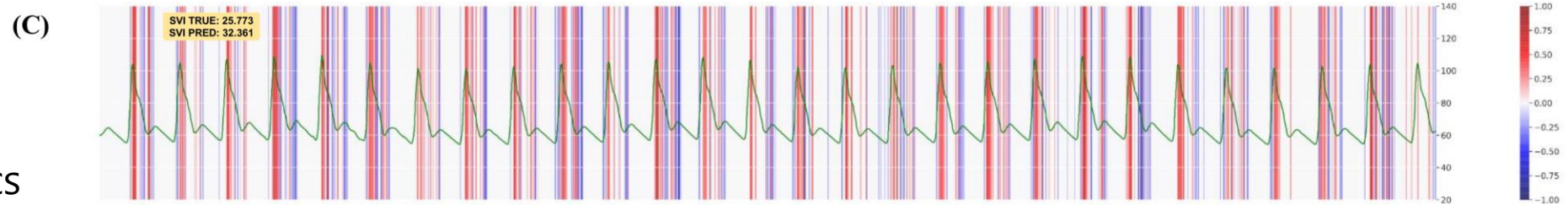
Low SV



High SV



Ablation:
inaccurate
demographics



Ablation:
Noisy a-line

